COURSE-BASED PROJECT DOCUMENTATION | DOC-10

Syntax Error Detection & Location

Location Precision, yylloc, Lookahead, & Formatting Scenarios

1. Syntax Error Detection Mechanism

A syntax error occurs when the next lookahead token cannot legally follow the grammar path for the current state. Bison halts normal reduction, triggers the callback void yyerror(const char *s), and sets syntax_error_occurred = 1.

2. Pinpoint Error-Line Extraction

```
void yyerror(const char *s) {
    syntax_error_occurred = 1;
    error_line = yylloc.first_line ? yylloc.first_line : yylineno;
}
```

Because YY_USER_ACTION in Flex continuously updates yylloc.first_line with the exact line number of every token, yyerror() captures the exact 1-indexed source line without drifting.

3. Standardized Output Format

- Success: SUCCESS: Valid Java Syntax (return code 0)
- Syntax Error: ERROR: Line <N>: Syntax error near '<TOKEN>' (return code 1)
- File I/O Error: ERROR: Could not open file '<PATH>' (return code 2)

4. Error Accuracy Verification Across Edge Cases

Scenario	Code Sample	Expected & Verified Line
Line 1 Error	class { }	Line 1: Syntax error near '{'
Line 2 Error	class T { int x = ; }	Line 2: Syntax error near ';'
After Blank Lines	5 blank lines followed by bad token	Exact line after blanks counted
After Block Comments	/* multi-line comment */ on lines 1-5	Line 6 error counted correctly
After Text Blocks	""" multi-line string """	Line after text block counte...
Missing Semicolon	int a = 10 int b = 20;	Line 2: Syntax error near 'i...